

GEOMETRICAL PACKING AND COVERING PROBLEMS

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INTRODUCTION

In this paper, we consider some packing and covering problems of a geometrical and usually recreational nature. Section 1 is on a *packing* problem. In Section 2, we consider a generalized type of *covering*, of the plane, by rectangles (§ 2.2 and 2.3) or polyominoes (§ 2.4).

Sections 3 and 4 are on *tilings*, also called partitions, dissections, and other names. In Section 3 we partition a rectangle. The four subsections are on fairly distinct ways of doing this. There are brief digressions on higher dimensions. In Section 4, we consider tilings of the plane, using polyominoes (§ 4.1) or arbitrary polygons (§ 4.2) as pieces.

The treatment is elementary; proofs are hardly given. The stress is on defining problem areas and pointing out open problems.

1. PACKING A SQUARE WITH UNIT SQUARES

Let $S(z)$ be a square with side z , let $n^*(z)$ be the maximum number of unit squares that can be packed into $S(z)$, and let $W(z) = z^2 - n^*(z)$.

ERDŐS & GRAHAM [5] have shown

$$(1) \quad W(z) = O(z^{7/11}) \quad (z \rightarrow \infty)$$

by a quite remarkable construction. One of the open problems they mention is to determine a non-trivial lower bound for W . Such a bound has been found by ROTH & VAUGHAN [19], who have proved that

$$W(z) \geq c(\|z\|z)^{\frac{1}{2}},$$

where $\|z\|$ is the distance from z to the nearest integer, and where $c = 10^{-100}$.

Since c is small, the result can be considered as an asymptotic one.

In [19], an unpublished result by Montgomery is mentioned, implying that the constant $7/11$ in (1) can be lowered to $(3-\sqrt{3})/2$.

We now consider fixed "small" values of z . Let $z^*(n)$ be the side of the smallest square into which n unit squares may be packed. Then obviously

$$(2) \quad \sqrt{n} \leq z^*(n) \leq \lceil \sqrt{n} \rceil.$$

The exact value of $z^*(n)$ is known only for $n = 2, 3, 5$ and the squares of integers^{*)}. In some of the remaining cases, the upper bound of (2) has been improved by suitable packings. (See table 1.) They are not difficult to reconstruct, except perhaps the packing for $n = 19$, which is shown in figure 1. I have not been able to improve on the upper bound in (2) for any n in a range $k^2 + k, \dots, (k+1)^2 - 1$, although it is obvious from (1) that such improvement is possible for n sufficiently large.

n	upper bound	n	upper bound
10	$3 + \frac{1}{2}\sqrt{2} \doteq 3.707$	37	$6 + \frac{1}{2}\sqrt{2} \doteq 6.707$
11	$\frac{5}{2} + \sqrt{2} \doteq 3.914$	38	"
17	$4 + \frac{1}{2}\sqrt{2} \doteq 4.707$	39	$4 + 2\sqrt{2} \doteq 6.828$
18	$2 + 2\sqrt{2} \doteq 4.828$	40	"
19 ¹⁾	$4 + \frac{2}{3}\sqrt{2} \doteq 4.943$	50	$7 + \frac{1}{2}\sqrt{2} \doteq 7.707$
26	$5 + \frac{1}{2}\sqrt{2} \doteq 5.707$	51	"
27	$5 + \frac{1}{2}\sqrt{2} \doteq 5.707$	52	"
28	$3 + 2\sqrt{2} \doteq 5.828$	65	$5 + \frac{5}{2}\sqrt{2} \doteq 8.536$

Table 1.

Best known upper bounds for $z^*(n)$

To demonstrate a technique for finding non-trivial lower bounds, we outline a proof of the following result (which implies $z^*(5) = 2 + \frac{1}{2}\sqrt{2}$).

^{*)} According to A. Schrijver, E. Bajmóczy of Budapest has shown that $z^*(7) = 3$.

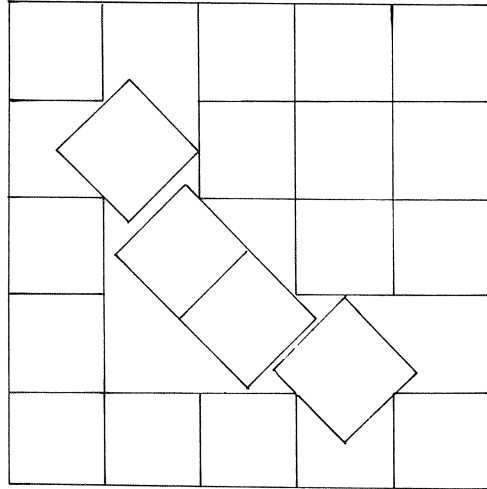


Figure 1.

A square packed with 19 unit square

PROPOSITION 1. $S' := S(2 + \frac{1}{2}\sqrt{2} - \epsilon)$ cannot be packed with 5 unit squares ($\epsilon > 0$).

OUTLINE OF PROOF. Take an S' and draw four lines in its interior, parallel to the sides and at a distance $1 - \epsilon/3$ from the sides (see figure 2). It is sufficient to show that any unit square $S(1)$ in S' covers at least one of the points A, B, C, D. There are 3 cases.

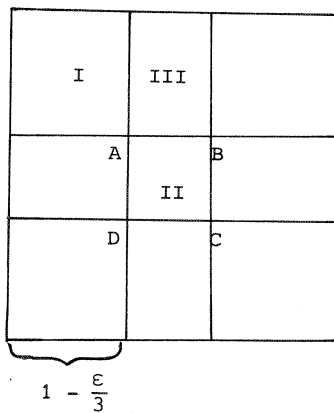


Figure 2.

- 1) The centre of $S(1)$ is in region I. Then an easy calculation in analytic geometry shows that A is covered.
- 2) The centre of $S(1)$ is in II. Suppose the centre M is closest to A. Then the distance $d(A, M)$ is $< \frac{1}{2}$, hence A is covered by $S(1)$.
- 3) The centre is in III. Without loss of generality we assume one vertex of $S(1)$ on the upper edge of S' . Again, a simple calculation shows that the length of the intersection of $S(1)$ and the line at distance $1 - \epsilon/3$ from the upper edge has length $> \frac{1}{2}\sqrt{2} - \epsilon/3$, hence A or B is covered. $\square$

2. GENERALIZED COVERINGS

2.1. Introduction

Let P and Q be polyominoes. Copies of P are placed on the square lattice, such that the sides are on lattice-lines, forming a *constellation of P* . A constellation of P is called *Q -saturated* if any copy of Q placed on the lattice such that its sides are on lattice-lines, has at least one square in common with some P .

If Q is the 1-omino, then a Q -saturated constellation of P is just a covering of the plane with P . This justifies the term "*generalized covering*".

The cases where P is the 1-omino, and Q is one of the pentominoes have been considered by GOLOMB [10].

We intend to consider other special cases viz. with $P = Q$. From now on we use the term "saturated" instead of " Q -saturated". In section 2.2, P is a rectangular polyomino; in section 2.4, P is an n -omino ($2 \leq n \leq 5$). In section 2.3 we consider a limiting case: $a \times b$ rectangles where a and b are real. In all cases, we are interested in generalized coverings with *minimal* density. In order to avoid technical problems concerning the existence of a density, we restrict our attention to periodic constellations.

2.2. Discrete rectangles

Let P be an $a \times b$ rectangle with $a \leq b$. The constellation of figure 3 shows that the minimum density $d^*(a,b)$ satisfies

$$(3) \quad d^*(a,b) \leq \frac{2ab}{(a+b-1)^2 + (2a-1)^2}.$$

On the other hand, a constellation of the type in figure 4 shows that

$$(4) \quad d^*(a,b) \leq \frac{2ab}{(2b-1)\min\{4a-2, a+b-1\}}.$$

A proof of (4) can be given as follows. If $b > 3a-2$, we choose $y = z = a-1$, and we obtain the upper bound

$$\frac{ab}{(2a-1)(2b-1)}.$$

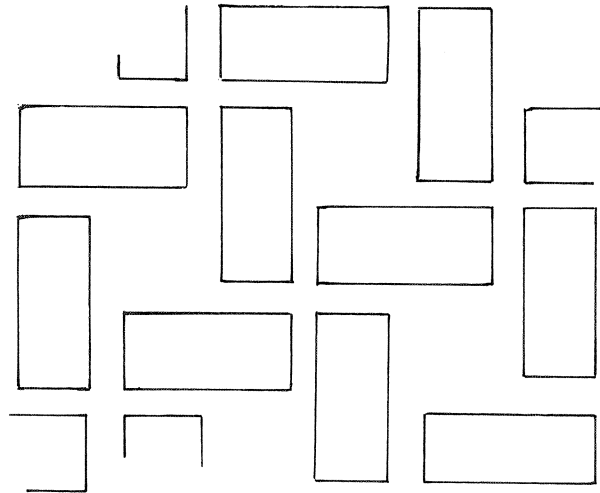


Figure 3.

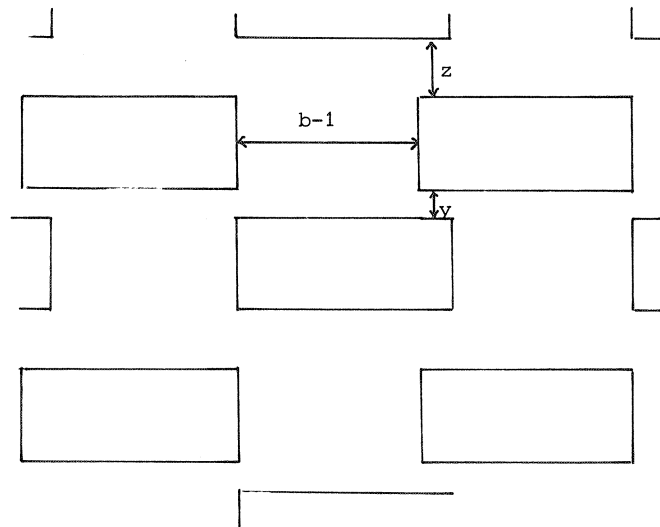


Figure 4.

If $b \leq 3a-2$, there are two cases: if $b-a$ is odd, we choose $y = z = (b-a-1)/2$, and if $b-a$ is even, we choose $y = (b-a-2)/2$, $z = (b-a)/2$. In both cases we obtain the upper bound

$$\frac{2ab}{(a+b-1)(2b-1)}.$$

Lower bounds for d^* can be obtained in several ways. The most successful method turned out to be the *shadow method* of Jagers; a detailed exposition is given in [14], a brief sketch in [9]. We state the following results of Jagers without proof.

$$(5) \quad d^*(a,b) \geq \frac{ab}{b(b-1)+3a(a-1)+1} \quad \text{if } b \leq 3a-1,$$

$$(6) \quad d^*(a,b) \geq \frac{2ab}{(a+b-1)(3a+b-2)} \quad \text{for all } a \leq b.$$

The lower bound (5) is better than (6) iff $b < 3a-1$. Combining the upper and lower bounds, we note that the minimal density $d^*(a,b)$ has been determined for $a = b$ and for $b = 3a-1$. In all other cases, the exact value of d^* is unknown. However, there is little doubt that the minimum is achieved for one of the types of constellations in figures 3 and 4.

2.3. Continuous rectangles

We replace the square lattice by a Cartesian coordinate system. Instead of $a \times b$ rectangles, we consider $\alpha \times 1$ rectangles with $0 < \alpha \leq 1$. We only allow positions of the rectangles in which the sides are parallel to the axes. The limits of the upper and lower bounds found so far are, in order of appearance:

$$(7) \quad d^*(\alpha) \leq \frac{2\alpha}{1+2\alpha+5\alpha^2} \quad (\text{from (3)}),$$

$$(8) \quad d^*(\alpha) \leq \frac{1}{4} \quad \text{for } \alpha \leq \frac{1}{3} \quad \left. \vphantom{\frac{1}{4}} \right\} \quad (\text{from (4)}),$$

$$(9) \quad d^*(\alpha) \leq \frac{\alpha}{1+\alpha} \quad \text{for } \alpha \geq \frac{1}{3}$$

$$(10) \quad d^*(\alpha) \geq \frac{\alpha}{1+3\alpha^2} \quad \text{for } \alpha \geq \frac{1}{3} \quad (\text{from (5)}),$$

$$(11) \quad d^*(\alpha) \geq \frac{2\alpha}{(1+\alpha)(1+3\alpha)} \quad (\text{from (6)}),$$

where $d^*(\alpha)$ is the minimum density in a saturated constellation of $\alpha \times 1$ rectangles. A pictorial summary is given in figure 5. The bracketed numbers refer to the above inequalities.

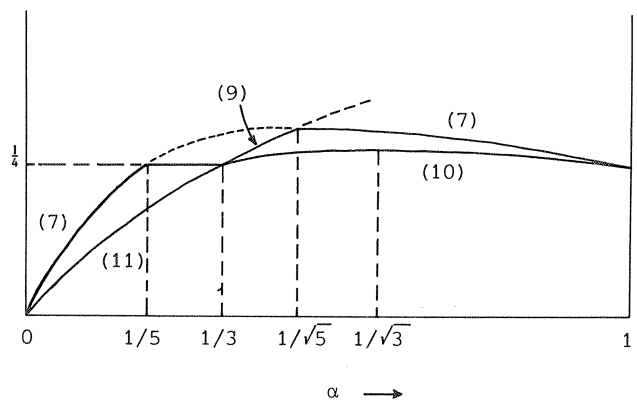


Figure 5.

The implicit conjecture at the end of section 2.2 has a continuous analogue: the upper bounds for $d^*(\alpha)$ given by (7), (8), (9) determine the minimum.

2.4. Polyominoes

In this section we consider generalized coverings with n -ominoes for $n \leq 5$.

Again, good upper bounds for $d^*(P)$, the minimum density of a saturated constellation of P 's, can be obtained from suitable constellations. Some of these appear in [9]. A summary of our best results is given in table 2. Most of the lower bounds have been obtained by the shadow method.

Note that there are some quite large ratios between upper and lower bounds.

The order (fifth column) is the number of polyominoes in an elementary cell or period parallelogram. For a definition of the symbols in the first column, we refer to [10].

P	lower bound	upper bound	ratio	order
I_2	2/3	2/3	1	1
I_3	1/2	3/5	1.20	2
L_3	6/11	6/11	1	2
I_4	2/5	8/17	1.18	2
L_4	32/77	1/2	1.20	1
O_4	4/9	4/9	1	1
T_4	4/9	4/9	1	2
Z_4	4/9	4/9	1	1
I_5	1/3	5/13	1.15	2
F_5	10/27	5/11	1.23	2
L_5	20/59	20/47	1.26	4
N_5	20/57	4/9	1.27	4
P_5	20/47	5/11	1.07	1
T_5	20/57	5/12	1.19	2
U_5	4/11	10/23	1.20	2
V_5	1/3	5/13	1.15	2
W_5	10/27	10/21	1.29	2
X_5	5/13	5/13	1	1
Y_5	20/57	4/9	1.27	4
Z_5	20/57	5/11	1.30	1 & 3

Table 2.

3. PARTITIONING A RECTANGLE

3.1. Different squares

A rectangle partitioned into different squares is called a *perfect (squared) rectangle*. It is called *compound* if it has a squared subrectangle, *simple* otherwise. The number of constituent squares is called the *order*.

The following short historical account is taken mainly from FEDERICO [6]. A more easily accessible account can be found in BONDY & MURTY [1].

The first perfect squared rectangle was published in 1925 by Morón; it is shown in figure 6. Note that it is simple.

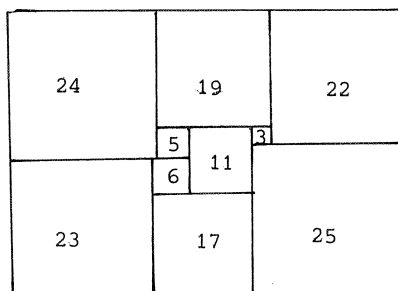


Figure 6.

The conjecture that no perfect square exists was defeated in 1939 by Sprague. He constructed a compound square of order 55. The first simple perfect square was published in 1940 by Brooks; its order is 55, too. If S is the smallest possible order of a simple perfect square, then Brooks' result implies $S \leq 55$. The subsequent history is as follows.

1940 Brooks, Smith, Stone, Tutte $S \geq 9$

1950 Brooks $S \leq 38$

Willcocks $S \leq 37$

1960 Bouwkamp, Duijvestijn, Medema $S \geq 15$

1962 Duijvestijn $S \geq 19$

1967 Wilson, Federico $S \leq 31$

Wilson $S \leq 25$

1977 Duijvestijn $S \geq 21$ [3].

Recently, on March 22, 1978 to be precise, Duijvestijn closed the gap by discovering a perfect simple square of order 21; for a description we refer to [4].

3.2. Congruent rectangles

For which P and Q can a $P \times Q$ -rectangle be partitioned into $r \times s$ -rectangles? Obviously, if P is a multiple of r or s , and Q is a multiple of the other one, then such a partition is possible. If $P = \lambda r + \mu s$ for non-negative integers λ and μ , while Q is a multiple of r and s , then again such a partition is possible. Of course, we may interchange P and Q here.

The following proposition, given by De BRUIJN [2], implies that there are no other solutions. His terminology is self-explanatory.

PROPOSITION 2. *If the box $A_1 \times \dots \times A_n$ can be filled with bricks $a_1 \times \dots \times a_n$, then at least one of the A_i is a multiple of a_1 , at least one of the A_i is a multiple of a_2 , etc.*

For a proof we refer to De Bruijn's article. Here we give a proof for $n = 2$, which is based on the same principle.

PROOF FOR $n = 2$. Colour the squares of the box (rectangle) with a_1 colours in a cyclic manner: let the colours be $0, \dots, a_1 - 1$ and assign two coordinates (x, y) to each square of the box ($0 \leq x \leq A_1 - 1$, $0 \leq y \leq A_2 - 1$), then assign the colour $x + y \pmod{a_1}$ to (x, y) .

Each small rectangle covers each of the colours a_2 times, whatever its position. On the other hand, if neither A_1 nor A_2 is a multiple of a_1 , e.g.

$A_1 = \lambda_1 a_1 + \mu_1$, $A_2 = \lambda_2 a_1 + \mu_2$ with $0 < \mu_1 < a_1$, then the number of occurrences of the colour $a_1 - 1$ in the upper-right $\mu_1 \times \mu_2$ rectangle is only $\max(0, \mu_1 + \mu_2 - a_1)$, which is less than the average $\mu_1 \mu_2 / a_1$. Hence A_1 or A_2 is divisible by a_1 .

In the same way one shows the divisibility by a_2 . $\square$

EXAMPLE. The box $6 \times 6 \times 6$ can not be filled with bricks of dimensions $1 \times 2 \times 4$.

We return to the n -dimensional case to quote another result from [2].

We call a brick $a_1 \times \dots \times a_n$ *harmonic* if the numbers $a_1, \dots, a_n$ can be rearranged to $a'_1, \dots, a'_n$ such that $a'_1 | a'_2$, $a'_2 | a'_3$, $\dots$, $a'_{n-1} | a'_n$.

PROPOSITION 3. *If a box $A_1 \times \dots \times A_n$ is filled with harmonic bricks $a_1 \times \dots \times a_n$ then there are integers $q_1, \dots, q_n$ such that $q_1 a_1, \dots, q_n a_n$ is a rearrangement of $A_1, \dots, A_n$.*

3.3. Tatami partitions

A partition of a $P \times Q$ rectangle into $r \times s$ rectangles is called a *Tatami partition* if $PQ > rs$ and if each $r \times s$ rectangle has the following property: the extension of each side either contains a side of the $P \times Q$ rectangle or has a point in common with the interior of an $r \times s$ rectangle. An example with $P = 5$, $Q = 6$, $r = 1$, $s = 2$ is given in figure 7.

PROPOSITION 4. *For each r and s with $r \neq s$, there exist numbers P and Q such that the $P \times Q$ rectangle has a Tatami partition into $r \times s$ rectangles.*

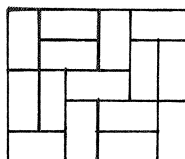


Figure 7.

OUTLINE OF PROOF. By a change of units, reduce to a case with $(r,s) = 1$. Enlarge figure 7 to a Tatami partition of $5rs \times 6rs$ into $rs \times 2rs$ rectangles. Next each $rs \times 2rs$ rectangle is subdivided into $r \times s$ rectangle as illustrated in figure 8 for the case $r = 2, s = 3$ (from left to right: 1 pile of vertical rectangles, r piles of horizontal, and finally $s - 1$ piles of vertical rectangles).

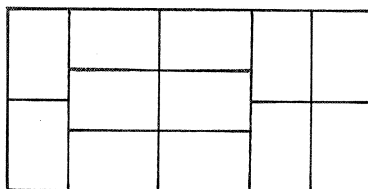


Figure 8.

It is now an easy matter to verify that a Tatami partition results. $\square$

The problem remains to determine, given r and s , which $P \times Q$ rectangles have a Tatami partition. In an unpublished report of 1965, R.L. Graham has given a complete solution. His result is as follows.

PROPOSITION 5. Let $(r,s) = 1$ and $PQ > rs$. Then a $P \times Q$ rectangle has a Tatami partition into $r \times s$ rectangles if and only if

- 1) r divides P or Q ; s divides P or Q ;
- 2) both P and Q have at least two representations in the form $xr+ys$ for positive integers x and y ;
- 3) $(P,Q) \neq (6,6)$ when $\{r,s\} = \{1,2\}$.

In the special case $r = 1$, WETTERLING [20] has independently obtained the following explicit result, which has a simple proof.

PROPOSITION 6. *The smallest $P \times Q$ rectangle which admits a Tatami partition into $1 \times s$ rectangles is the $(2s+1) \times 3s$ rectangle.*

OUTLINE OF PROOF. Suppose $P \times Q$ has a Tatami partition into $1 \times s$ rectangles. It is not difficult to show that this implies $\min(P, Q) \geq 2s+1$. On the other hand, from proposition 2 we know that P or Q is divisible by s . Hence the smallest candidate is the $(2s+1) \times 3s$ rectangle. To complete the proof, it is sufficient to give a Tatami partition for this case. We refer to figures 7 and 9; the latter gives the construction for $s = 4$. The generalization to arbitrary s is obvious. $\square$

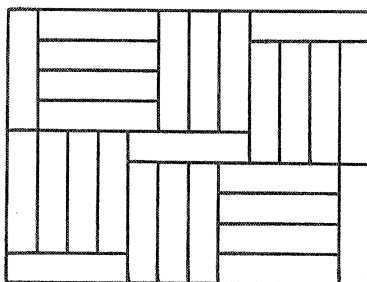


Figure 9.

3.4. Congruent polyominoes

Given a polyomino P , let $\mathcal{B}(P)$ be the class of rectangles which can be partitioned into copies of P . The first question is: "Is $\mathcal{B}(P)$ empty?" Second question: "If not, which rectangles belong to $\mathcal{B}(P)$?"

We start with a simple example. Let $P = L_3$. It is obvious that $2 \times 3 \in \mathcal{B}(L_3)$. Hence, all rectangles which can be partitioned into 2×3 rectangles, belong to $\mathcal{B}(L_3)$ (cf. § 3.2). Does $\mathcal{B}(L_3)$ contain other elements? Yes, $5 \times 9 \in \mathcal{B}(L_3)$, as is easily verified.

Hence, each rectangle which can be partitioned into 2×3 's and 5×9 's belongs to $\mathcal{B}(L_3)$. It is easily shown that $\mathcal{B}(L_3)$ contains no other rectangles. So we have a satisfactory description of $\mathcal{B}(L_3)$, and we might consider 2×3

and 5×9 as its *prime* elements.

KLARNER [16] showed that there are, for each polyomino P , only a finite number of prime rectangles. When I attempted to generalize his proof to d -dimensional polyominoes, I made an error (cf. [17]), but Klarner succeeded in finding a correct proof for the d -dimensional case [18].

The complete set $\mathcal{P}(P)$ of prime rectangles is known only in a relatively small number of cases, although for certain polyominoes much partial information is available.

Since [17] has been written, the following results have been obtained. HASELGROVE [13] has found a Y_5 -partition of the 15×15 rectangle, thereby solving an old problem, viz. "Does any odd number of Y_5 's tile a rectangle?"

KLARNER [18] has determined $\mathcal{P}(P_8)$ (see figure 10), it consists of the 4×4 , 5×16 , 6×8 , and 7×16 rectangles.

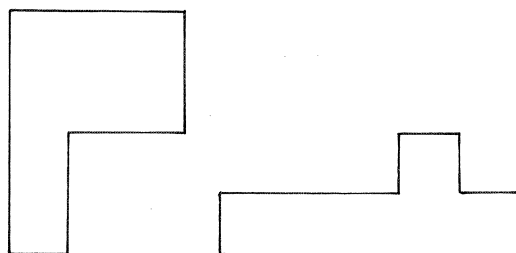


Figure 10, P_8 and Y_6 .

As far as I know, it is still not known whether Y_6 packs any rectangle.

In 3 dimensions, much more can be done. For example, the tetracube Z_4 fills boxes of sizes $2 \times 3 \times 4$, $2 \times 4 \times 4$, $2 \times 4 \times 5$. Less obvious examples are T_5 which fills $3 \times 10 \times 10$, and F_5 which fills $4 \times 5 \times 10$. It is not known whether W_5 or Z_5 fills any box.

4. TILING THE PLANE

4.1. Polyominoes

If a polyomino does not tile a rectangle, it may still tile the plane, as the example Z_4 shows. In fact, Z_4 tiles a strip of width 2, hence the plane. GOLOMB [11] has considered this phenomenon in more detail. A polyomino

may tile a rectangle (R), a strip (S), a bent strip (BS), a half-strip (HS), a quadrant (Q), a half-plane (HP), the plane (P), or "all" shapes in the empty collection (N). Golomb proves that there is a hierarchy between these shapes, as shown in figure 11.

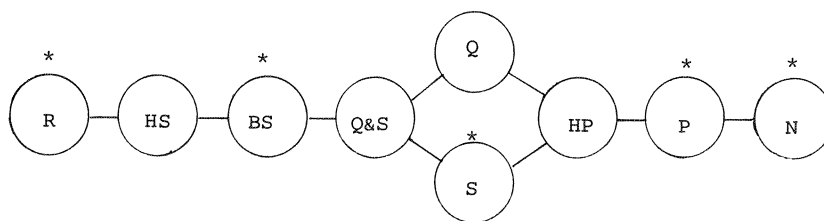


Figure 11.

For example, if a polyomino tiles a half-strip, it tiles a bent-strip, etc. Each polyomino has its place in the hierarchy, in the sense that it tiles the corresponding shape X, but not a shape which is higher in the hierarchy (i.e. further to the left in figure 11). We say the polyomino is *characteristic* for the shape X. However, only for the starred places in figure 11, it has been possible to determine characteristic polyominoes.

In a later paper [12], Golomb has determined characteristic sets of polyominoes for each of the shapes.

GARDNER [8B] reports on an interesting sufficient condition for a polyomino to tile the plane. It is due to CONWAY. The formulation below is due to Doris SCHATTSCHEIDER [21].

THEOREM. A tile T tiles the plane if there are six consecutive points $v_1, \dots, v_6$ at least three distinct on the boundary of T which satisfy the following conditions:

- (i) the boundary segment $[v_1, v_2]$ is congruent to $[v_5, v_4]$ by a translation τ in which $\tau(v_1) = v_5$ and $\tau(v_2) = v_4$;
- (ii) $[v_2, v_3]$, $[v_3, v_4]$, $[v_5, v_6]$, $[v_6, v_1]$ are centro-symmetric.

The conditions are illustrated by the polyomino of figure 12. The usefulness of the criterion becomes clear when applied to the 108 heptominoes. It turns out that 101 of them satisfy the criterion, so that only 7 cases have to be considered separately. Four of these are non-tilers.

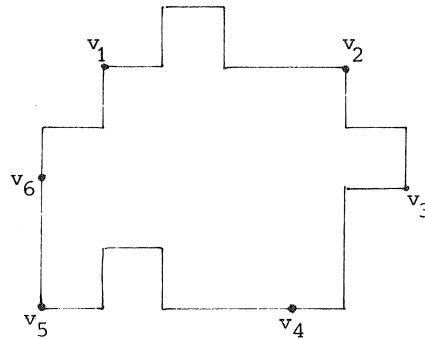


Figure 12.

It has been shown by GOLOMB [12] that no finite algorithm exists which decides whether copies from a finite set of polyominoes tile the plane. If the set contains only one element, the decidability question is open.

But even when a polyomino is known to tile the plane, many questions can be asked, e.g. "In what ways does it tile the plane?"

In figure 13 we indicate three ways of tiling the plane with copies of C_6 ; in figure 14 we present a much more complicated tiling with an elementary cell containing 32 copies of C_6 .

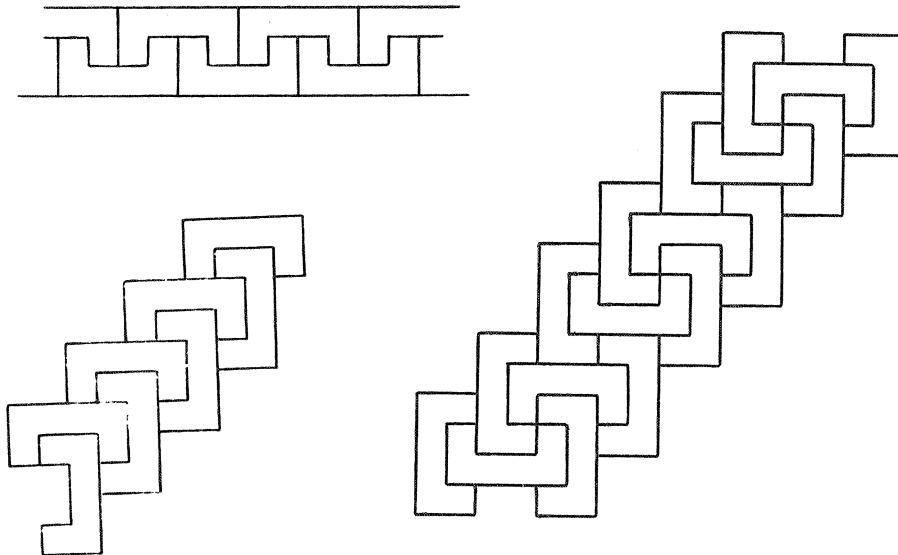


Figure 13.

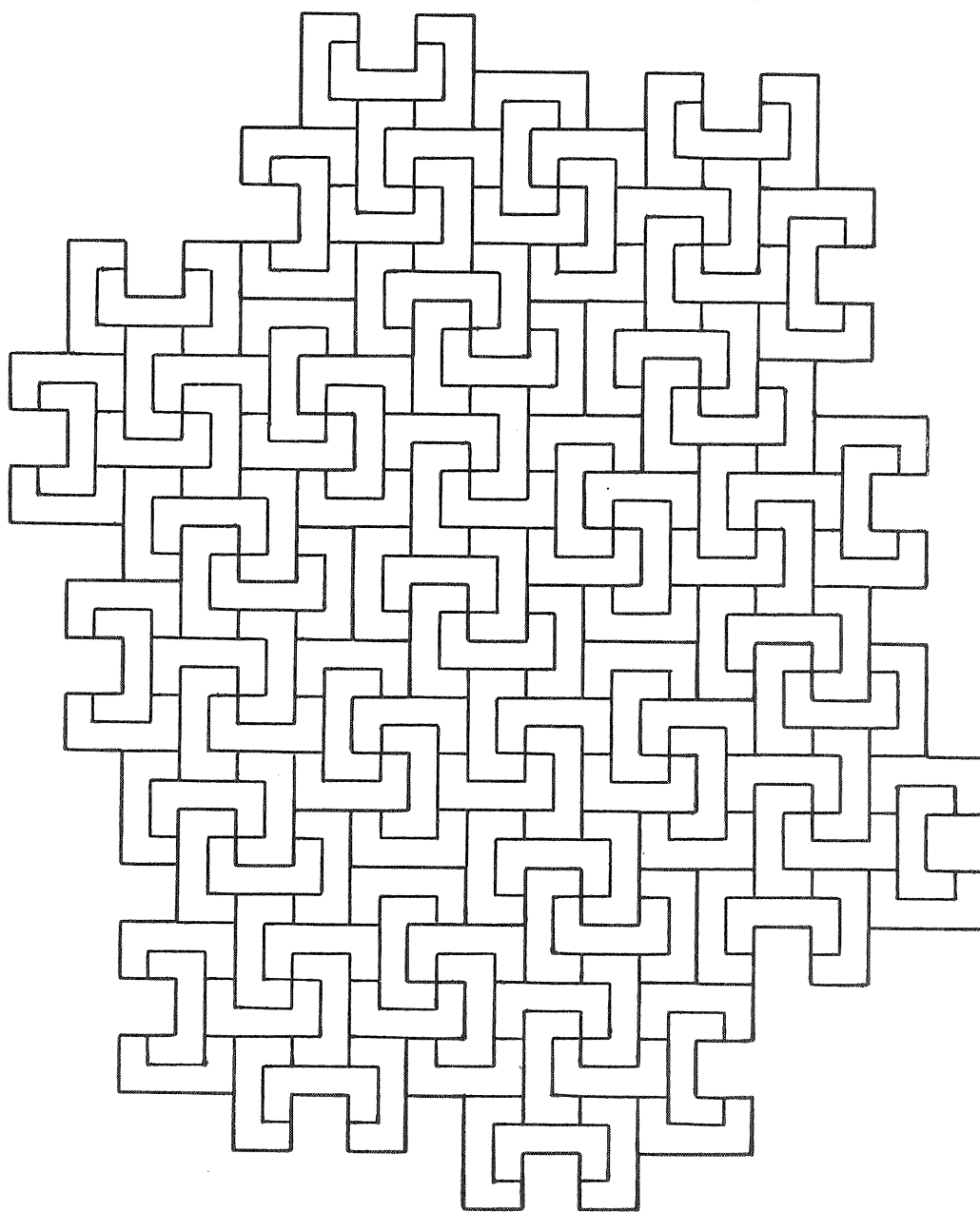


Figure 14.

According to GARDNER [8B], A.W. Bell has discovered 19 types of tilings with L_4 .

4.2. Tiling the plane with congruent polygons

It is easily seen that each triangle and each quadrangle, convex or not, tiles the plane. Curiously, it seems that the question as to what *types* of tilings are possible with quadrangles, say, has not been considered at all.

With pentagons or polygons of higher orders, it is possible to tile the plane in special cases only. For the moment, we restrict our attention to *convex* n -gons ($n \geq 5$).

The case $n = 5$ has a romantic history. In 1918, five types of pentagons which tile the plane were discovered by K. Reinhardt. To illustrate, we describe "type 2" in Kershner's notation [15]. Let the vertices be called A, B, C, D, E in cyclic order, and let $EA=a$, $AB=b$, $BC=c$, $CD=d$, $DE=e$. Then a pentagon of type 2 is a pentagon with $A+B+D=2\pi$, $a=d$. In 1968, KERSHNER [15] published 3 new types. He claimed completeness, but did not give the proof, for reasons of space. In 1975, GARDNER [8A] wrote about Kershner's results in the Scientific American, and after a couple of months, he published a new type [8C], found by R.E. James. An example is shown in figure 15. The requirements are $A=90^\circ$, $C+D=270^\circ$, $2D+E=2C+B=360^\circ$, $a=b+c+e$. It is clear that the case of the convex pentagons is not closed.

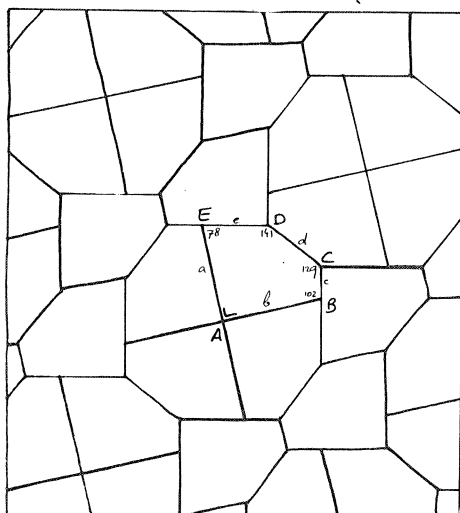


Figure 15.

For hexagons, the situation is much simpler: there are three types, all found by Reinhardt. A tiling with congruent convex n -gons is not possible when $n \geq 7$.

We return to not necessarily convex n -gons to quote from FEJES TÓTH's book [7]: "The general tiling problem consists of obtaining a description of all partitions of the plane into equal (but not necessarily equivalent) parts. The difficulty inherent in this problem (brought into prominence by Hilbert) is illustrated by the very interesting partition due to Voderberg (1936, 1937)". A figure showing that partition can be found not only in [7], but also in GARDNER's column [8D]. The latter describes a very simple way to obtain Voderberg's partition, found by Golomb. He starts with a non-periodic triangle tiling like the one in figure 16a. He then slides the "upper half" to the left to obtain figure 16b. Finally, the lateral sides of the triangles are "crooked" to yield something like Voderberg's 9-gons (figure 17).

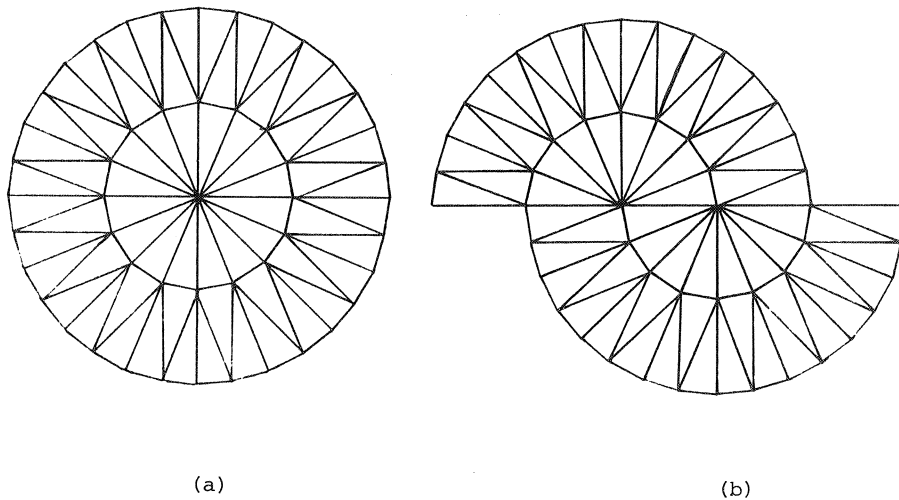


Figure 16.

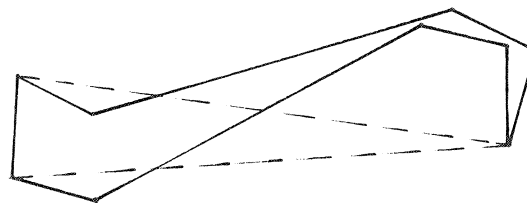


Figure 17.

The tilings of figure 16 are non-periodic. Obviously, the triangle admits periodic tilings as well. An open question is whether any polygon exists, which tiles the plane non-periodically only.

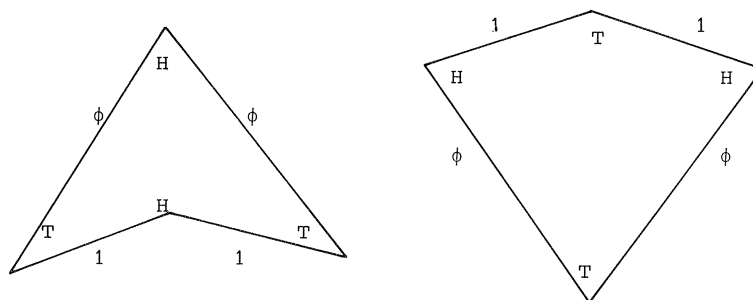


Figure 18.

According to GARDNER [8D], R. Berger has constructed a set of more than 20,000 cells, copies of which tile the plane non-periodically only. He also reports on the present record: Penrose has discovered the set of 2 polygons shown in figure 18, which tile the plane non-periodically only. The letters H and T near the vertices are intended as restrictions: two pieces may only touch at equal letters. The sides have lengths 1 and ϕ , where $\phi = \frac{1}{2} + \frac{1}{2}\sqrt{5}$; the angles are all multiples of 36° . Gardner mentions several properties of Penrose's polygons, e.g. in each tiling the ratio of the number of "kites" to "darts" is ϕ . Also, there are uncountably many different tilings. However, each pair of tilings has arbitrarily large finite areas in common! For further details, we refer to Gardner's article. No proofs are given (with the exception of one incomplete proof), but the article is beautifully illustrated.

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Note added at the Second printing (1982).

- ¹⁾ The upper bound for $z^*(19)$ has been improved by Charles F. Cottingham, and independently by Jan Kok to $3+4\sqrt{2}/3 \doteq 4.886$. See also Gardner's column in the October and November 1979 issues of the Scientific American. Jaap van der Woude has shown that $z^*(41) \leq 2+7\sqrt{2}/2 \doteq 6.950$.